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Project Title:	Confocal microscopy of the colloidal glass transition		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
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Comments:			
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Task Description:	How is a piece of glass different from other solid materials? As a glassy material is cooled from the molten, liquid state, it gradually flows slower and slower and at some point is considered a glass, in contrast with regular phase transitions from a liquid to a solid which happen in well-defined circumstances. To study the transition to a glass state, we conduct ground-based experiments with concentrated colloidal suspensions, a model system which has a glass transition. These are suspensions of solid particles in a liquid, and when the particle concentration is high enough, the material is glassy and behaves like a paste. By using a confocal microscope to follow the three-dimensional motion of colloidal particles, we can directly observe the microscopic behavior responsible for the macroscopic viscosity divergence of glasses. We conduct a variety of experiments to understand the properties of the glassy state, and to probe the sample as the glassy state is approached. Our largest set of experiments use small magnetic particles to locally “poke” on the colloidal samples. By using an external magnet, we can pull a magnetic particle at constant force; oscillate a magnetic particle back and forth; and rotate clusters of magnetic particles. The response to these perturbations changes significantly near the glass transition. For example, we study the motion of the colloidal particles surrounding a moving magnetic particle. We find the magnitudes of displacements of the surrounding particles decay exponentially in space, in contrast to the flow expected for a regular fluid. We study how this decay length varies with geometry (translation or rotation), packing fraction, and the applied force. Our results address broader issues of a universal description of the origin of the glass transition. To use direct imaging to quantify the microscopic behavior of particles as the colloidal glass transition is approached. Specifically, both structural properties and dynamical properties are measured and compared with theoretical descriptions of the glass transition. We also seek to characterize the aging of colloidal glasses; we wish to determine how colloidal glasses change over time. The overarching goal is to clarify the mechanisms underlying the glass transition and to provide key data to test various theories of the glass transition. Confocal microscopy of the colloidal suspensions allows us to both directly follow the motion of individual particles and to visualize structural arrangements as the transition is approached. These measurements are crucial to testing and extending theoretical understanding of both dense colloidal suspensions and glasses in general. Moreover, the insight from these experiments will be useful to the processing of industrially important materials. Finally, by understanding how the glass transition occurs in these model systems and how the dynamics are affected by the system details, we may be able to design novel materials with specific properties.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In this year of funding, we continued our experiments on perturbing dense colloidal suspensions with magnetic beads. We are continuing to write up our results, mostly involving work by the PI and Prof. Piotr Habdas (St. Joseph's University) who was previously funded by this grant as a postdoctoral researcher. Additionally, the experimental work was continued by three undergraduates and one graduate student. Dense colloidal suspensions are a simple model system, which becomes glassy when the colloidal particle concentration is above a critical threshold. Below this threshold, the sample is analogous to a liquid, and colloidal particles move around the sample due to Brownian motion. The glass transition is identified at the point where the diffusion of colloidal particles through the sample can no longer be observed on reasonable experimental time scales. The goal of our experiments is to observe how such samples respond to being locally perturbed in a variety of ways. In particular, we wish to characterize how the response changes in liquid-like samples as the transition is approached. We have investigated how the sample responds when it is stirred by a tiny magnetic stir bar. We place a very small concentration of 5 micron diameter magnetic particles into the sample of 2 micron diameter PMMA colloidal particles. Occasionally, the magnetic particles stick together, forming dimers. We put a permanent magnet on a stepper motor above the sample, and by rotating the magnet can rotate the dimer and stir the sample locally. Our preliminary data indicate that there is a stirring region of finite radius. Dr. Habdas is analyzing this data to determine how the radius of the perturbed region changes as the colloidal concentration is increased; we have hints that this radius grows. Ben Bluth and Jason Hay, undergraduates, are examining what happens when a magnetic bead is pulled with a large constant force. We published data on this in 2004 for a particular colloidal suspension where the magnetic bead was twice the diameter of the colloidal particles. Hay's experiments examine the force/velocity relationship when we vary the size ratio between the magnetic bead and the colloidal particles. Bluth's experiments are studying what happens in a binary mixture of two particle sizes, where crystallization will not be a problem. Thus, these samples will allow us to conduct very long duration experiments, previously prevented by crystallization. In particular we'll be able to study magnetic beads moving at extremely slow velocities. Their preliminary data indicate that we get similar results to our earlier work. Lov Goel, another undergraduate, is studying the motion of the particles surrounding the translating magnetic bead. Similar to Habdas's studies of the rotating dimers, we have an indication that the size of the disturbance grows as we approach the colloidal glass transition. Goel is carefully characterizing this and also measuring how this disturbance size depends on the velocity of the magnetic bead. Gianguido Cianci (a graduate student) is studying the aging of colloidal glasses. By stirring a dense sample vigorously, the sample is initialized, and the subsequent dynamics show aging: the time scale for particle motion grows with the age of the sample (the time since stirring ended). In our first year of funding, we studied the aging in monodisperse colloidal samples (resulting in one publication so far). Currently, Cianci is studying how the structure varies as the sample ages. All of the methods we have tried to quantify the structure show that the system has the same structure as it ages; we have not found a structural signature of aging. However, Cianci does find that the local structure influences which particles move -- and it is precisely the motion of particles that age the sample. Thus, the structure does influence the aging even if we have not yet seen how aging influences the structure. Dr. Vikram Prasad (a postdoctoral fellow) studies the Brownian motion of colloidal particles at an interface. In three dimensions, the motion of particles can provide information about the bulk rheological properties of the sample, using the techniques of microrheology. For example, if the particle motion is diffusive then the bulk viscosity can be determined by measuring the diffusion constant. In Dr. Prasad’s experiments, the particles are constrained to move in 2D, and thus their motion probes the 2-dimensional surface viscosity, which is set by the surfactant concentration at the interface. However, the interface is between two 3D reservoirs of fluid (the oil and the water), so the motion is also influenced by their viscosities. We study the spatially correlated motion of pairs of particles, and find that their behavior

	changes from 3D-influenced to primarily 2D-influenced as the surface viscosity rises. We believe these are the first experiments to probe this type of behavioral transition.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Papers from Meeting Proceedings	Cianci GC, Courtland RE, Weeks ER. "Invariance of structure in an aging colloidal glass." The Second International Conference on Flow Dynamics, Sendai (Japan), 16-18 November 2005. FLOW DYNAMICS: The Second International Conference on Flow Dynamics, eds. M. Tokuyama and S. Maruyama (AIP Conference Series, vol. 832), p. 21-25. http://dx.doi.org/10.1063/1.2204459 , May-2006
Presentation	Weeks, E. R. "Local perturbations of colloidal glasses" 3rd Workshop on Complex Systems, Sendai, Japan Nov-2005
Presentation	Weeks, E. R. "Aging in a colloidal glass" Relaxations in Complex Systems, Lille, France Jul-2005
Presentation	Weeks, E. R. "Particle Tracking" 5th World Congress on Particle Technology, Orlando, FL Apr-2006